

Week 4 — Confounding: What to Adjust For, and Why

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: obesity and death in MASLD (simplified from Kueh et al. 2026)

Today's question: is obesity protective?

Among adults with MASLD, people **with obesity** died **less often** than people without it: **7.9%** vs **13.0%**. The hazard ratio (HR) for death, obese vs not obese:

What the model adjusts for	HR (95% CI)
Nothing (crude)	0.67 (0.56–0.81)
Age and sex	0.91 (0.75–1.09)
The paper's Model 1 : age, sex + diabetes, hypertension, cholesterol, kidney disease, heart attack, cancer, medicines	0.77 (0.64–0.93)
The paper's Model 2 : Model 1 + smoking, sedentary time	0.80 (0.66–0.97)

Same 5,911 people in every row. $HR < 1$ = a lower death rate with obesity.

Our running example

- ▶ **Who:** US adults (NHANES 2007–2018) with **MASLD** — fatty liver (fatty liver index, **FLI** ≥ 60) plus at least one cardiometabolic risk factor
- ▶ **Exposure (A): obesity** — BMI ≥ 30 (≥ 25 for Asian participants). **Outcome (Y):** death from any cause, compared with a Cox model's HR
- ▶ **People:** 5,911 with complete data — 4,620 with obesity, 1,291 without
- ▶ **Adjusting for (or conditioning on)** a variable = comparing people with the **same value** of it — in a model, by stratifying, by restricting the sample, or by **selecting** who enters the study. **L** = measured covariates; **U** = unmeasured ones

Simplified for today. The paper crosses BMI with waist size to make four groups. We use **BMI alone**, which the paper does not report — today's numbers are our own reproduction.

Read more · Kueh et al., *BMJ Open* 2026 · EpiMethods Notation and glossary · our code: `lectures/analysis/Week4_bmi_exposure.R`

By the end of today you can...

1. Say what **causal question** an adjusted estimate answers, and what must be true for it to answer it
2. Use a **DAG** (a diagram of what causes what) to choose what to adjust for
3. Name the **role** a variable plays — confounder, mediator, collider, instrument, proxy
4. Explain why **odds ratios and hazard ratios** change when you adjust, even with no confounding
5. Report adjusted results **without the Table 2 fallacy**

Read more · EpiMethods *Concepts*, “Causal roles”: ehsanx.github.io/EpiMethods/confounding0.html

Part 1 · What question are we asking?

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EpiMethods *Concepts*: “The Epistemological Divide” · “The Counterfactual Framework”

Explaining or predicting?

Predicting: *“Which adults with MASLD are most likely to die?”*

Anything that tracks death helps — **including** diabetes caused by obesity.

Same R code (+ diabetes), **opposite rules**. The paper uses causal words (*“confers the poorest survival”*) but adjusts like a prediction model — for diabetes, kidney disease and heart attack, which obesity may itself cause.

Read more · EpiMethods *Concepts*, “The Epistemological Divide: Explanatory versus Predictive Modeling”

Explaining: *“Would they live longer if they were not obese?”*

Adjusting for diabetes **caused by** obesity hides part of the effect we want.

What we want vs what we can see

Person	Obese?	Y(1): if obese	Y(0): if not obese
Ana	yes	died	?
Ben	no	?	alive

- ▶ Everyone has two **potential outcomes**; we only ever see one. So we aim for an average: the **average treatment effect**, $ATE = E[Y(1) - Y(0)]$
- ▶ The data give $E[Y | A = 1] - E[Y | A = 0]$: 7.9% – 13.0%. The two match only if the groups are **exchangeable** — alike except for obesity
- ▶ Here they are not: people without obesity are older (**54.6** vs **50.5** years), more often men and more often smokers (**24.7%** vs **18.0%**)

Our hope is **conditional exchangeability**: comparable within levels of L (same age, sex ...). It **cannot be tested** — a DAG writes down why we believe it.

Read more · EpiMethods *Concepts*, “The Counterfactual Framework” · video · Hernán & Robins, *What If*, ch. 1–2 (free online)

Positivity, consistency and a tricky exposure

- ▶ **Positivity** — is it violated here? **No**. It needs every kind of person to appear in both groups, and they do: given age, sex, race/ethnicity, smoking and activity, each person's predicted chance of obesity lies between **35%** and **97%** — never 0% or 100%
- ▶ **But it is thin in places:** among women aged 18–39, **671** have obesity and only **53** do not, so what we conclude about young women rests on 53 people
- ▶ **Consistency:** “not obese” must mean **one** thing. It doesn't — diet, exercise, surgery, a drug, smoking or cancer can each make someone not obese, with different effects on death. *What If* uses exactly this example, down to asking whether we mean “*a massive liposuction*”

Positivity ✓ holds (thin for young women). **Consistency** × is doubtful: “obesity” is not one well-defined thing we could change.

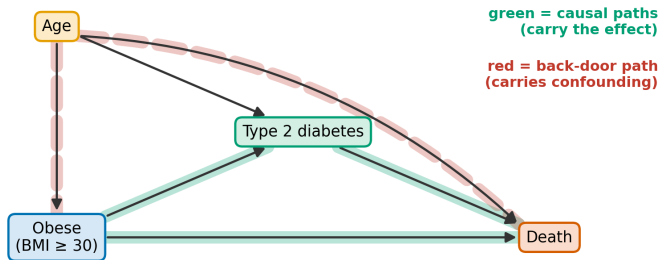
Read more · Hernán & Robins, *What If*, ch. 3: “Consistency: first, define the counterfactual outcome” · Hernán & Taubman, *Int J Obes* 2008

Part 2 · Drawing our assumptions: DAGs

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EpiMethods *Concepts*: “Structural and Knowledge-Based Selection Techniques”

A DAG, and the two kinds of path in it

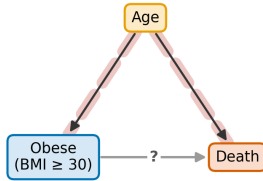


- ▶ **Arrow** = “directly causes”; **no arrow** = a claim of no direct effect
- ▶ **Causal paths** (green) follow the arrows forward: obesity → death; obesity → diabetes → death.
Back-door paths (red) start with an arrow **into** obesity: obesity ← age → death — they carry **confounding**

Goal: block every back door, keep the causal paths open, open nothing new — here, adjust for **age**.
dagitty.net finds the adjustment sets for you.

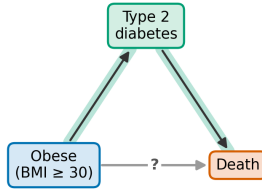
Three building blocks

Fork — confounding



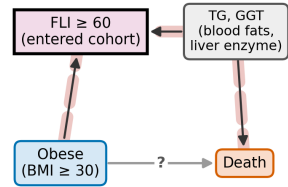
Path OPEN → adjusting CLOSES it

Chain — mediation



Path OPEN and carries the effect
→ adjusting REMOVES part of it

Collider — selection



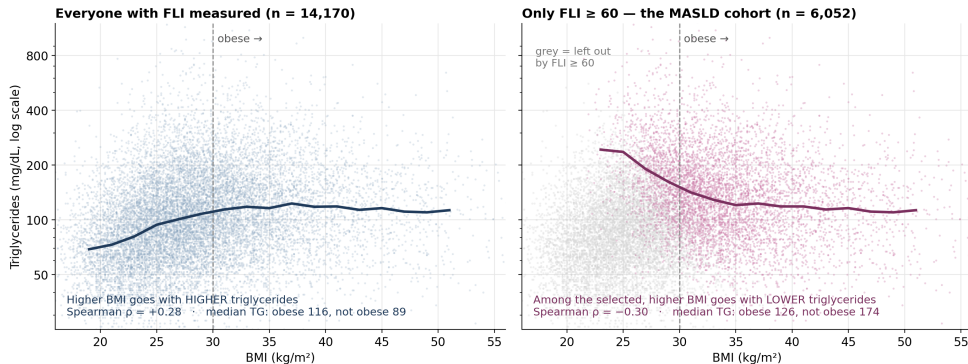
Path CLOSED → conditioning OPENS it

Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

A **collider** (two arrows collide into it) blocks its path on its own. Adjusting for it — or for something it causes — **opens** the path.

Read more · EpiMethods *Concepts*, “The Three Elementary Causal Structures” · tutorials: confounding, mediator, collider

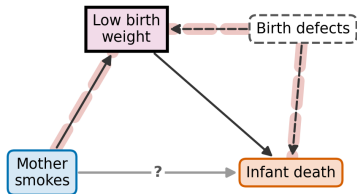
Selecting on $FLI \geq 60$: a collider you can see



FLI rises with BMI **and** with triglycerides (TG). Keep only $FLI \geq 60$, and people with a low BMI stay in **only** if their TG is high: $\rho = +0.28$ among everyone, **-0.30** among the selected.

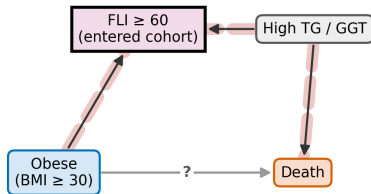
Selection can make obesity look protective

The birth-weight paradox



Studying only low-birth-weight babies = conditioning on it

The obesity paradox in this MASLD cohort



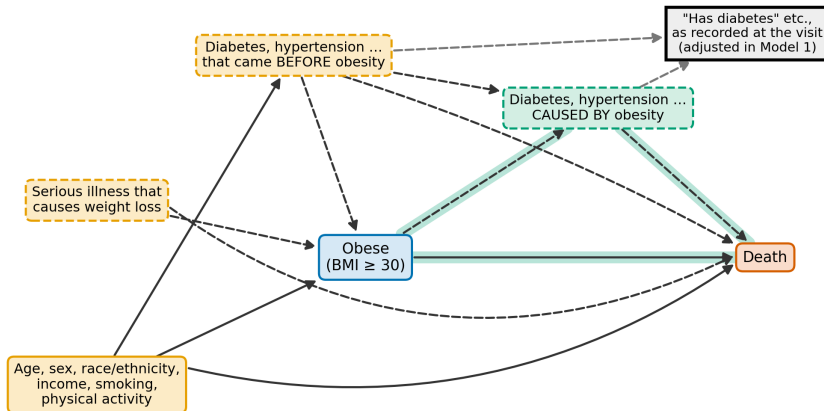
Studying only people with FLI ≥ 60 = conditioning on it

Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

People without obesity reach $FLI \geq 60$ with higher **TG** (174 vs 126 mg/dL) and **GGT** (36 vs 25 U/L). If these raise the risk of death, obesity looks **protective** — **index-event bias**. The birth-weight paradox (left) is the classic case.

Describing ✓ people with obesity died less often here. **Causal** × “obesity protects” is not supported.

A working DAG for obesity and death



Serious illness (unmeasured) causes weight loss *and* death — reverse causation. **"Has diabetes"** is recorded once, so it mixes disease before obesity (a confounder) with disease caused by it (a mediator).
dagitty: **no set of the variables we measured** is valid.

Reading the moves with the DAG

Step	HR	What was added	What the DAG predicts
Crude	0.67	—	confounded
+ age, sex	0.91	confounders	closing back doors → toward 1 ✓
Model 1	0.77	the recorded conditions	partly mediators → away from 1 ✓
Model 2	0.80	+ smoking , sedentary time	smokers are leaner and die more: a confounder → toward 1 ✓
Pre- exposure only	0.95 (0.78–1.14)	age, sex, race/ethnicity, smoking, sedentary time	the closest measured set to the DAG's

Every move goes the way the DAG predicts — **support, not proof**.

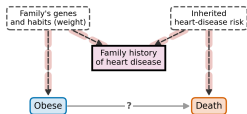
Four more roles a variable can play

Instrument



Causes the exposure only → don't adjust

M-bias — a collider measured before exposure



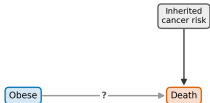
Adjusting it opens the red path (arrows drawn for illustration)

Proxy of an unmeasured confounder



Adjusting income PARTLY closes the red path

Outcome-only cause



Not a confounder: adjusting adds no bias

- ▶ **Instrument** (causes the exposure only): **don't** adjust — it can amplify hidden bias
- ▶ **M-bias variable** (a collider measured before the exposure): **don't**, if that is all it is. If it might also confound, **adjust** — M-bias tends to be small
- ▶ **Proxy** for an unmeasured confounder: **adjust** — it helps, but doesn't fully fix
- ▶ **Outcome-only cause**: optional — no bias, often a sharper estimate

Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

Read more · EpiMethods Z-bias tutorial · VanderWeele, "Principles of confounder selection", *Eur J Epidemiol* 2019

Can't draw the whole DAG? Rules of thumb

Rule	Adjust for	Catch
Pre-treatment	everything measured before the exposure	may include instruments and M-bias variables
Common cause	only causes of both	too strict — drops real confounders
Disjunctive cause	pre-exposure causes of the exposure or the outcome	you only need to know one arrow
Modified disjunctive cause	the same, minus instruments, plus proxies	a practical default ✓

Every rule needs **time order**, and one NHANES visit can't tell whether diabetes came before obesity.

Read more · EpiMethods *Concepts*, “Empirical Criteria for DAG-Deficient Scenarios” · video · VanderWeele & Shpitser, *Biometrics* 2011

Part 3 · Why data-driven shortcuts fail

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EpiMethods *Concepts*: “Modelling criteria for variable selection”

Table 1 describes; a DAG chooses

	Obese	Not obese	p-value	SMD
Number of people	4,620	1,291		
Died, n (%)	366 (7.9%)	168 (13.0%)	<0.001	0.17
Age, mean (SD)	50.5 (16.1)	54.6 (15.5)	<0.001	0.26
Women	52.6%	27.7%	<0.001	0.52
Current smokers	18.0%	24.7%	<0.001	0.16
Type 2 diabetes	26.2%	20.9%	<0.001	0.13

- ▶ **Every p-value is <0.001** — with 5,911 people, even small gaps are “significant”. The **SMD** (standardised mean difference) measures the gap's **size**: at EpiMethods' cut-off of 0.2, sex (0.52) and age (0.26) are imbalanced; smoking (0.16) and diabetes (0.13) are not
- ▶ A gap says a variable **differs** between the groups — not whether it is a confounder, a mediator or neither. That takes the DAG

Change-in-estimate: “if it moves, keep it”

The rule: add a variable; keep it if the exposure’s estimate changes by more than 10%.

- ▶ Age and sex: **+34%** → “confounders” ✓ right
- ▶ The recorded conditions: **−15%** → “confounders” × probably partly mediators
- ▶ Smoking and sedentary time: **+4%** → “not confounders” × smoking **is** a confounder

An estimate can move because confounding was removed, **or** part of the effect was removed (mediators), **or** bias was added (colliders), **or** because ORs and HRs move with no confounding at all (after the next part). The rule can’t tell which.

Read more · EpiMethods *Concepts*, “Change-in-Estimate” · tutorial · video

p-values, stepwise and machine learning

- ▶ **Confounding is not a significance test:** a strong confounder can be “not significant” in a small sample
- ▶ **Stepwise selection** keeps whatever predicts death best — here, diabetes, kidney disease, heart attack — and its confidence intervals ignore the search: **too narrow**
- ▶ **Purposeful selection** is more careful, but still blind to causal structure: it keeps colliders, because colliders move the estimate
- ▶ **Machine learning** (e.g., LASSO) is excellent at prediction; in causal work, use it only for the prediction parts of a method (propensity scores, later in the course)

Choose confounders from **knowledge** — a DAG and the rules of thumb — not from the data.

Read more · EpiMethods *Concepts*, “Statistical Significance (Stepwise Selection)”, “Purposeful Selection of Covariates”, “Machine Learning (ML)” · Heinze, Wallisch & Dunkler, *Biom J* 2018

Part 4 · What adjustment does to the effect measure

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EpiMethods *Concepts*: “Collapsibility and the Choice of Effect Measure”

Collapsibility: ORs move without confounding

*Made-up numbers. Half of each age group is exposed, so age is **not** a confounder. OR = odds ratio · RR = risk ratio · RD = risk difference.*

Age group (people)	Risk, exposed	Risk, unexposed	OR	RR	RD
60 or older (200)	0.80	0.50	4.00	1.60	0.30
Under 60 (200)	0.50	0.20	4.00	2.50	0.30
Everyone (400)	0.65	0.35	3.45	1.86	0.30

- ▶ **RD** and **RR** are **collapsible**: the overall value is an average of the subgroups'
- ▶ **OR** is not: 4.00 in **both** age groups, 3.45 overall. **HRs behave like ORs**
- ▶ Adjusting for age moves the OR **away from 1** (3.45 → 4.00) — change-in-estimate would wrongly call age a confounder

Read more · EpiMethods collapsibility tutorial · Naimi & Whitcomb, *Am J Epidemiol* 2020

Conditional vs marginal — both are right

- ▶ **Conditional:** among people with the **same covariate values** — what a regression coefficient gives. Here, **OR = 4.00**
- ▶ **Marginal:** in the **whole population** — everyone exposed vs everyone unexposed, what a randomised trial gives. Here, **OR = 3.45**
- ▶ The gap is **not bias**. It needs a real effect, and it always pushes the conditional OR or HR **away from 1**

Here the marginal OR equals the crude OR **only because age is not a confounder**. With confounding, the marginal effect is the **standardised** one (next slide) — not the crude.

Read more · EpiMethods collapsibility tutorial · Hernán, “The hazards of hazard ratios”, *Epidemiology* 2010

Getting a marginal effect: g-computation

1. Fit an outcome model with the exposure and the covariates
2. Set **everyone to obese**; predict each person's risk and average
3. Set **everyone to not obese**; predict and average again — then compare

```
# fit = glm(died ~ obese + age + female + ..., family = binomial, data = d)
d1 <- transform(d, obese = 1); d0 <- transform(d, obese = 0)
p1 <- mean(predict(fit, newdata = d1, type = "response"))
p0 <- mean(predict(fit, newdata = d0, type = "response"))
c(RD = p1 - p0, RR = p1 / p0, OR = (p1 / (1 - p1)) / (p0 / (1 - p0)))
```

One logistic model gives the **marginal** RD, RR and OR together. For survival, the same idea gives a **standardised survival curve**: everyone obese vs no one obese.

Read more · EpiMethods collapsibility tutorial, “Marginal RD, RR and OR” · Snowden, Rose & Mortimer, *Am J Epidemiol* 2011 · Naimi & Whitcomb, *Am J Epidemiol* 2020

Part 5 · Reporting what you estimated

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EpiMethods *Concepts*: “The Table 2 Fallacy”

Table 2 fallacy: one model, many questions

One Cox model, as a paper's Table 2 would print it:

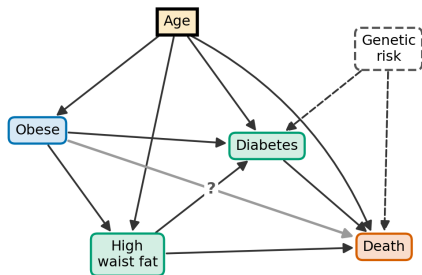
In one model: obesity + high waist fat + age + sex + diabetes	HR (95% CI)
Obese	0.80 (0.65–0.99)
High waist fat (WHtR $\geq$ 0.6)	1.17 (0.87–1.57)
Age, per 10 years	2.30 (2.12–2.49)
Type 2 diabetes	1.65 (1.39–1.97)

Table 2 fallacy = reading every row as that variable's own effect. Each row answers a **different** question — and each question needs **its own DAG and adjustment set**.

Read more · EpiMethods *Concepts*, “Why A Single Model Fails” · video · Westreich & Greenland, *Am J Epidemiol* 2013

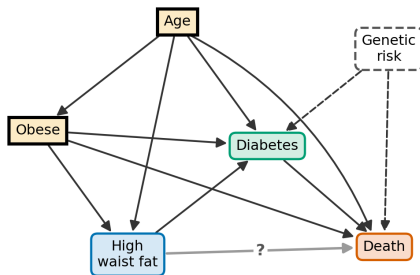
Four questions, four adjustment sets (1 of 2)

Q1 · Effect of obesity



Adjust: age. Not waist or diabetes — both are on obesity's path

Q2 · Effect of high waist fat



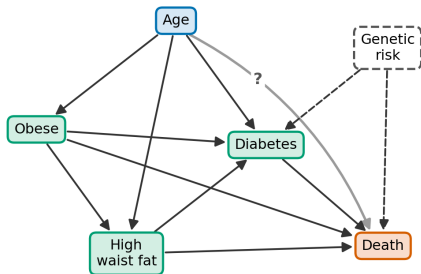
Adjust: age AND obesity (a confounder here). Not diabetes

Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

Same DAG, different question — different roles. Waist fat is a **mediator** in Q1 but the exposure in Q2; obesity is the exposure in Q1 but a **confounder** in Q2.

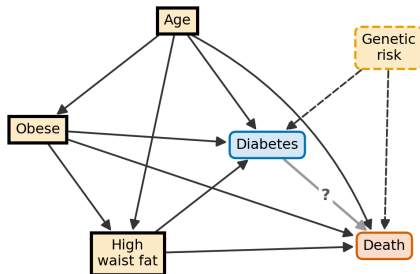
Four questions, four adjustment sets (2 of 2)

Q3 · Effect of age



Adjust: nothing — nothing in this DAG causes age

Q4 · Effect of diabetes



Adjust: age, obesity, waist — AND genetic risk (unmeasured)

Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

Age needs no adjustment — nothing causes it. **Diabetes** needs a variable nobody measured (genetic risk), so no model in our data answers Q4.

The one model vs each question's own model

Question	Its own DAG says adjust for	Its own model	The one model's row
Q1 · obesity	age (+ sex, race/ethnicity, smoking, activity)	0.95 (0.78–1.14)	0.80 (0.65–0.99)
Q2 · high waist fat	age, obesity (+ the same)	1.24 (0.93–1.67)	1.17 (0.87–1.57)
Q3 · age, per 10 years	nothing	2.38 (2.21–2.56)	2.30 (2.12–2.49)
Q4 · diabetes	age, obesity, waist + genetic risk (unmeasured)	1.71 (1.43–2.04)*	1.65 (1.39–1.97)

- ▶ **Obesity** looks “protective” (0.80) in the one model because it holds waist fat and diabetes fixed — both on obesity's path. Its own model says **0.95**
- ▶ Even where the numbers are close, each row **means** something different

Rule: one exposure, one model, one row reported — covariates in a footnote. * Still confounded: genetic risk is not measured.

Key takeaways

1. An adjusted estimate answers a causal question only under **assumptions** — draw them as a **DAG**
2. Adjust for **confounders**; not for **mediators** (for the total effect), **colliders** or **instruments**.
Selecting who is in the study counts as adjusting
3. Choose confounders from knowledge, not p-values or change-in-estimate
4. **ORs and HRs** move on adjustment even without confounding; RDs and RRs don't
5. **One exposure per model** — the other rows are not effects
6. Obesity's apparent protection (**0.67**) mostly disappears with pre-exposure variables (**0.95**).
Adjustment can't fix what's unmeasured (serious illness) or the selection ($FLI \geq 60$)

Next: Week 5 — the paper's four BMI $\times$ waist groups (effect modification). **Exit question:** the one arrow in your own paper's DAG you are least sure of?

Where to read more

Topic	EpiMethods	Key reading
Notation, positivity, consistency	Notation and glossary	Hernán & Robins, <i>What If</i> (free)
Potential outcomes	<i>Concepts</i> — Counterfactual Framework · video	Hernán & Robins, <i>What If</i> , ch. 1
DAGs	<i>Concepts</i> — Grammar of causal diagrams · videos 1 2 3	Greenland et al. 1999; Tennant et al. 2021
Roles, by simulation	tutorials 1 2 3 3b 4	—
Rules of thumb	<i>Concepts</i> — Empirical Criteria · video	VanderWeele 2019
Data-driven selection	<i>Concepts</i> — Modelling criteria · tutorial	Heinze et al. 2018
Collapsibility, standardisation	tutorial · marginal RD/RR/OR	Naimi & Whitcomb 2020
Table 2 fallacy	<i>Concepts</i> — Why a single model fails · video	Westreich & Greenland 2013
Test yourself	EpiMethods quiz	—

References (1 of 2)

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- ▶ Hernán MA. The hazards of hazard ratios. *Epidemiology* 2010.
- ▶ Hernán MA, Hernández-Díaz S, Robins JM. A structural approach to selection bias. *Epidemiology* 2004.
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- ▶ Hernán MA, Taubman SL. Does obesity shorten life? The importance of well-defined interventions to answer causal questions. *Int J Obes* 2008.
- ▶ Hernández-Díaz S, Schisterman EF, Hernán MA. The birth weight “paradox” uncovered? *Am J Epidemiol* 2006.

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- ▶ Kueh MTW, et al. Body weight categories and fat distribution in relation to all-cause mortality among adults with metabolic dysfunction-associated steatotic liver disease. *BMJ Open* 2026;16:e113719. doi:10.1136/bmjopen-2025-113719
- ▶ Naimi AI, Whitcomb BW. Estimating risk ratios and risk differences using regression. *Am J Epidemiol* 2020.
- ▶ Snowden JM, Rose S, Mortimer KM. Implementation of G-computation on a simulated data set: demonstration of a causal inference technique. *Am J Epidemiol* 2011.
- ▶ Tennant PWG, et al. Use of directed acyclic graphs (DAGs) to identify confounders in applied health research: review and recommendations. *Int J Epidemiol* 2021.
- ▶ VanderWeele TJ. Principles of confounder selection. *Eur J Epidemiol* 2019.
- ▶ VanderWeele TJ, Shpitser I. A new criterion for confounder selection. *Biometrics* 2011.
- ▶ Westreich D, Greenland S. The Table 2 fallacy: presenting and interpreting confounder and modifier coefficients. *Am J Epidemiol* 2013.

Appendix: our working DAG as dagitty code

Paste into the *Model code* box at dagitty.net. With Ill, Before and After marked latent (never measured), dagitty reports that **no adjustment set exists**. Remove the latent tags from Ill and Before, and it returns **{Before, Ill, L}**.

```
dag {  
  Obese [exposure]  Death [outcome]  Ill [latent]  Before [latent]  After [latent]  
  L -> Obese  L -> Death  L -> Before  
  Ill -> Obese  Ill -> Death  
  Before -> Obese  Before -> Death  Before -> After  
  Obese -> After  After -> Death  Obese -> Death  
  Before -> Blend  After -> Blend  
}
```

L = age, sex, race/ethnicity, income, smoking, physical activity. Ill = serious illness that causes weight loss. Blend = the recorded “has diabetes (etc.)” variable.